ZCAS UNIVERSITY
SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

2025 DEFERRED EXAMINATIONS

OBJECT ORIENTED PROGRAMMING EXAM (CCS 2121)

DURATION: 3 HOURS MARKS: 100

INSTRUCTIONS

1. This question paper contains **6** printed pages.
2. An answer booklet is provided for all your work.
3. Clearly write your particulars in the spaces provided in the answer booklet.
4. Do not open the paper until you are told to do so.
5. There are **SIX (6)** questions in this paper, answer **ALL** Questions in **SECTION A** and any **TWO (2) in SECTION B.**
6. **All** Questions carry equal marks.
7. Marks will be awarded for showing all your workings.

.....

SECTION A: COMPULSARY QUESTIONS (Answer All)

Question One

i) Describe the two types Casting and demonstrate using the java code how they work [4 marks]

ii) Design a program using algorithm to calculate both CA and final examination mark when given student marks .The weighting for the CA are: 10 marks theory and 30 marks practical making 40% CA and examination mark 60% [4 marks]

iii) Write a Java program to get input from the user [4 marks]

iv) Explain the four main concepts of object oriented programming [4 marks]

v) With the help of diagram, explain the steps involved in executing java program starting from Source code to execution. [4 Marks]

Total 20 marks

Question Two

a) The following Java program contains bugs will prevent it from running. Identify these bugs and explain why the program won't run. [8 Marks]

```
public class University {
    private String universityName;

    public University(String name) {
        universityName = name
    }

    public class Department {
        private String departmentName;

        Department(String name) {
            this.departmentName = name;
        }

        void printDepartmentInfo() {
            System.out.println("Department: " + departmentName);
        }

        public class Staff {
            private String name;
            private String position;

            public Staff(String name, String title) {
                this.name = name;
                this.position = title;
            }

            void printStaffInfo() {
                System.out.println("Staff: " + name + " working as " +
position;
            }
        }

        public void offerCourse(String courseName) {
            class Course {
                private String courseTitle;

                Course(String title) {
                    courseTitle = name;
                }

                void showCourse() {
                    System.out.println("Course Offered: " + courseTitle + "
in department " + this.departmentName);
                }
            }

            Course c = new Course(courseName);
            c.showCourse();
        }

        // Anonymous inner class example
        public void announceEvent() {
            Runnable announcer = new Runnable() {
                public void run() {
                    System.out.println("Welcome to " +
departmentName.toUpper();
                }
            };
            announcer.run()
        }
```

b) Rewrite the above program to make it run smoothly in any IDE of your choice. [12 Marks]

Total 20 marks

Question Three

a) The Department of Environmental Conservation has contracted you to build a WildLife Monitoring System that tracks different types of animals in conservation parks [10 Marks].

You are required to:

i. Design an abstract class Animal with common behaviors like eat() and makeSound(), both abstract.
ii. Create a subclass Mammal which extends Animal, providing a default eat() behavior.
iii. Create a specific class Lion that extends Mammal and overrides both eat() and makeSound() with lion-specific behavior.
iv. Add a method hunt() in the Lion class to simulate hunting logic.
v. Also create an Elephant class (subclass of Mammal) which overrides makeSound().

b) A smart home company called HomeVerse is building a new system to manage different types of smart devices. Each device may support different capabilities such as turning on/off, adjusting brightness, or connecting to the internet. [10 Marks]

Your software must:

- Create three methods:
  i. Switchable with turnOn() and turnOff()
  ii. Dimmable with setBrightness(int level)
  iii. Connectable with connectToWiFi(String ssid)

Total 20 marks

SECTION B: ANSWER ANY TWO QUESTIONS

Question Four

i) With the help suitable diagrams , contrast the two types of program testing black box and white box testing [4 marks]

ii) Describe the two main categories of data types giving text book examples [4 marks]

iii)Using appropriate diagram, describe the architecture of Java Virtual Machine (JVM) [4 marks]

vi) Outline stages involved in developing computer program [4 marks]

v) Computers are good at handling repetitive tasks using loops than humans. With the help of appropriate syntax for each type of loop, describe the following types of loops

For Loop

Do/ while

While/Loop [4 marks]

Total 20 marks

Question Five

i) Define Debugging [4 marks]

ii) Explain the following types of program testing:
Unit testing, integrated testing and Acceptance testing [12 marks]

iii) Why do you think it is critical to test the program to avoid the problem of "*flying by wire*" as the case was with Ethiopian airline which killed a number of passengers [4 marks]

Total 20 marks

Question Six

CampusSmart is a newly launched digital system designed to help universities streamline the management of academic and non-academic resources such as classrooms, staff, students, and schedules. The system aims to serve as a digital backbone to various departments, offering an efficient way to manage assets, handle class allocations, and track faculty workload. The university administration has hired your software development team to build the first version of this system using object-oriented programming principles.

The initial module will focus on three main entities: FacultyMember, Classroom, and Course. Every faculty member has a unique ID, name, designation (e.g., Lecturer, Professor), and a list of assigned courses. Each course has a course code, title, credit hours, and is linked to a classroom where it will be taught. Classrooms, in turn, have room numbers, capacity, and a type (e.g., Lab, Lecture Hall). You are to design a system where each faculty member can be assigned to teach multiple courses, each course should be scheduled in a specific classroom, and changes should be tracked via logs.

The solution should demonstrate object-oriented design principles such as encapsulation (to protect internal states of objects), composition (e.g., a course has a classroom), and inheritance (e.g., both FullTimeFaculty and PartTimeFaculty may extend FacultyMember but have different workload rules). There should also be a logging mechanism to track every assignment action (e.g., when a faculty is assigned to a new course). This logging can be implemented using a static utility class.

Your system should allow adding new faculty members using constructors, assigning them courses using methods, changing classroom assignments, and calculating total faculty workload in terms of credit hours. Bonus functionality would include automatically suggesting a classroom for a course based on availability and capacity. All data should be displayed cleanly via toString() methods and grouped using arrays or ArrayList.

i) Design a base class FacultyMember with common attributes and methods. Create two subclasses: FullTimeFaculty and PartTimeFaculty. Override the calculateWorkload() method in each subclass where Full-time faculty must teach at least 12 credit hours. [10 Marks]

ii) Create a class Classroom with attributes like roomNumber, capacity, and type. Then, create a Course class that takes a Classroom object in its constructor, along with course attributes. In the `FacultyMember` class, write overloaded methods `assignCourse()` and One method that accepts just a `Course` object, and another that accepts a course and a boolean `forceAssign`, which bypasses workload limit (e.g., admin override). [10 Marks]

Total 20 marks

END OF EXAMINATION